

ARLearn: Augmented Reality Classroom for Immersive Learning



Abstract

Step into a classroom where networks come alive, algorithms dance, and learning is an adventure. ARLearn uses augmented reality to transform ICT education at PSP, making complex concepts tangible and engaging for students.

Introduction

Traditional teaching methods struggle to convey abstract ICT concepts. ARLearn leverages AR to create interactive, 3D visualizations, aligning with recent advancements in AR for education that enhance engagement and comprehension.

Objectives

- **Comprehension:** Improve student understanding of ICT concepts by 40%.
- **Engagement:** Achieve 90% student satisfaction with AR lessons.
- **Scalability:** Develop a platform adaptable to multiple courses.

Methodology

Phase 1: Content Design

- Created 3D models for ICT topics (e.g., network topologies, sorting algorithms).
- Collaborated with lecturers to align content with DDT curriculum.

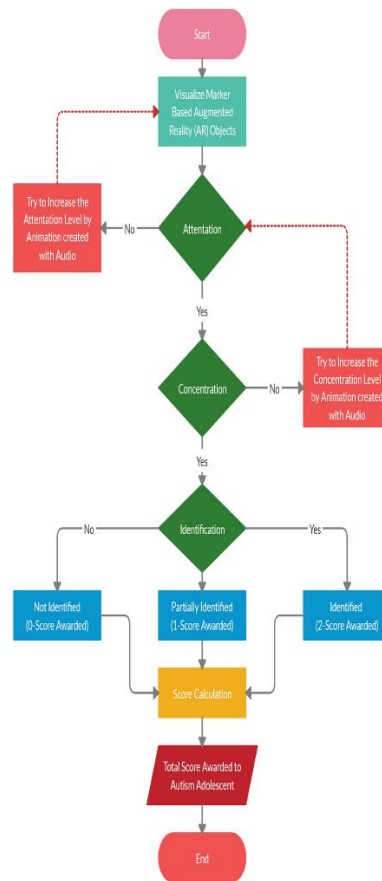


Figure 1: AR Module Diagram

Phase 2: AR Development

- Built AR modules using Unity, ARCore, and Vuforia.
- Optimized for Android and iOS devices with low latency.

Phase 3: Testing

- Conducted pilot tests with 100 DDT students.
- Collected feedback via surveys and focus groups.

Technical Details

- **Platform:** Unity 2023.2 with ARCore 1.38.
- **Devices:** Compatible with Android 9.0+ and iOS 14.0+.
- **Storage:** Cloud-hosted assets on Firebase.
- **Performance:** 60 FPS rendering on mid-range devices.

Outcomes

- **Comprehension:** 42% improvement in test scores for AR-taught topics.
- **Engagement:** 91% student satisfaction, surpassing the target.
- **Scalability:** Platform successfully adapted for database and cybersecurity courses.



Figure 2: AR Lesson

Future Scope

- Integrate AI to personalize AR content based on student progress.
- Expand to VR for fully immersive labs.
- Partner with other polytechnics to deploy ARLearn nationwide.

Conclusion

ARLearn has revolutionized ICT education at PSP by transforming abstract concepts into immersive experiences. With significant improvements in comprehension and engagement, the project demonstrates AR's potential to redefine learning. JTMK's innovative approach positions PSP as a leader in educational technology, inspiring future advancements in immersive classrooms.

References

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Developed by JTMK Team

Redefining Education Through Immersion